Technology Stack

Date	29 September 2026
Team ID	SWTID-2026-6564
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	*ServiceNow Service Portal / Employee Center and the standard Incident form (HTML, CSS, client scripts)*	Employees already use it to raise tickets, so no separate app is needed. It is responsive and works in a browser on any device.
2	**Backend / Server-Side**	*ServiceNow Flow Designer (trigger: Incident Created, conditions and update actions) with server-side JavaScript for keyword matching where needed*	Flow Designer is a low-code, visual tool on the platform, so the classification logic is easy to build, read and change. Scripts handle only the parts that need custom logic.
3	**Database / Data Storage**	*ServiceNow platform database: Incident table, custom Classification Rules table, User Group table*	Tables are built into the platform with relationships, ACL security and reporting. Keeping keywords in a table means rules can be edited without touching the flow.
4	**Cloud / Hosting / Deployment**	*ServiceNow cloud instance (Personal Developer Instance for the project)*	Fully managed cloud hosting with no servers to set up, and it is free for learning and development.
5	**Version Control & CI/CD**	*ServiceNow Update Sets and Flow Designer versioning, with the Automated Test Framework; GitHub via Studio if needed*	Update Sets move the flow between instances, flow versions allow rollback, and automated tests check that classification still works after changes.

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6	**Third-Party APIs / Other Tools**	*ServiceNow email notifications, Reports and Dashboards (no external API required)*	Notifies the caller and assignment group, and shows ticket volume by category and group. Everything stays within the platform.

Summary

The whole solution is built on the ServiceNow platform. An employee creates an incident in the portal, a Flow Designer flow classifies and routes it using keyword rules stored in a table, and notifications and dashboards are provided by built-in platform features. Using one platform keeps the solution simple to build, secure and maintain, and needs no separate hosting or external services.